



Acute Bacterial Meningitis

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Acute bacterial meningitis is rapidly progressive bacterial infection of the meninges and subarachnoid space. Findings typically include headache, fever, and nuchal rigidity. Diagnosis is by cerebrospinal fluid (CSF) analysis. Treatment is with antibiotics and corticosteroids given as soon as possible.

(See also [Overview of Meningitis](#) and [Neonatal Bacterial Meningitis](#).)

Pathophysiology of Acute Bacterial Meningitis

Most commonly, bacteria reach the subarachnoid space and meninges via hematogenous spread. Bacteria may also reach the meninges from nearby infected structures or through a congenital or acquired defect in the skull or spine (see [Route of entry](#)).

Because white blood cells (WBCs), immunoglobulins, and complement are normally sparse or absent from cerebrospinal fluid (CSF), bacteria initially multiply without causing inflammation. Later, bacteria release endotoxins, teichoic acid, and other substances that trigger an inflammatory response with mediators such as WBCs and tumor necrosis factor (TNF). Typically in CSF, levels of protein increase, and because bacteria consume glucose and because less glucose is transported into the CSF, glucose levels decrease. Brain parenchyma is typically affected by inflammatory responses in acute bacterial meningitis.

Inflammation in the subarachnoid space may be accompanied by cortical encephalitis and ventriculitis.

Complications of bacterial meningitis are common and may include

- [Hydrocephalus](#) (in some patients)
- Arterial or venous infarcts due to inflammation and thrombosis of arteries and veins in superficial and sometimes deep areas of brain
- [Abducens palsy](#) due to compression or inflammation of the 6th cranial nerve
- Deafness due to inflammation of the 8th cranial nerve or structures in the middle ear

- [Subdural empyema](#)
- Increased intracranial pressure (ICP) due to cerebral edema
- [Brain abscess](#) (if the infection penetrates into brain parenchyma)
- [Brain herniation](#) (the most common cause of death during the acute stages)
- Systemic complications (which are sometimes fatal), such as [septic shock](#), [disseminated intravascular coagulation](#) (DIC), or hyponatremia due to [syndrome of inappropriate antidiuretic hormone secretion](#) (SIADH)

Etiology of Acute Bacterial Meningitis

Likely causes of bacterial meningitis depend on

- Patient age
- Route of entry
- Immune status of the patient

Age

In **neonates and young infants**, the most common causes of bacterial meningitis are

- Group B streptococci, particularly *Streptococcus agalactiae*
- *Escherichia (E.) coli* and other gram-negative bacteria
- *Listeria monocytogenes*

In **older infants, children, and young adults**, the most common causes of bacterial meningitis are

- *Neisseria meningitidis*
- *Streptococcus pneumoniae*

N. meningitidis meningitis occasionally causes death within hours. Sepsis caused by *N. meningitidis* sometimes results in coagulopathy and bilateral adrenal hemorrhagic infarction (Waterhouse-Friderichsen syndrome).

Haemophilus influenzae type B, previously the most common cause of meningitis in children < 6 years and overall, is now a rare cause in the United States and Western Europe, where the *H. influenzae* vaccine is widely used. However, in areas where the vaccine is not widely used, *H. influenzae* is still a common cause, particularly in children aged 2 months to 6 years.

In **middle-aged and in older adults**, the most common cause of bacterial meningitis is

- *S. pneumoniae*

Less commonly, *N. meningitidis* causes meningitis in middle-aged and older adults. As host defenses decline with age, patients may develop meningitis due to *L. monocytogenes* or gram-negative bacteria.

In **people of all ages**, *Staphylococcus aureus* occasionally causes meningitis.

TABLE

Causes of Bacterial Meningitis by Patient Age

Age Group	Bacteria
Neonates and young infants	Group B streptococci, particularly <i>Streptococcus agalactiae</i> <i>Escherichia (E.) coli</i> <i>Listeria monocytogenes</i> , <i>Staphylococcus aureus</i>
Older infants, children, and young adults	<i>Neisseria meningitidis</i> <i>Streptococcus pneumoniae</i> <i>Staphylococcus aureus</i> * <i>Haemophilus influenzae</i> (rare in developed countries but still seen in countries where the <i>H. influenzae</i> type B vaccine is not widely used)
Middle-aged adults	<i>S. pneumoniae</i> <i>S. aureus</i> * <i>N. meningitidis</i> (less common in this age group)
Older adults	<i>S. pneumoniae</i> <i>S. aureus</i> * <i>L. monocytogenes</i> Gram-negative bacteria
* <i>S. aureus</i> occasionally causes severe meningitis in patients of all ages. It is the most common cause of meningitis that develops after a penetrating head wound.	

Route of entry

Routes of entry include the following:

- By hematogenous spread (the most common route)
- From infected structures in or around the head (eg, sinuses, middle ear, mastoid process), sometimes associated with a CSF leak
- Through a penetrating head wound

- After a neurosurgical procedure (eg, if a ventricular shunt becomes infected)
- Through congenital or acquired defects in the skull or spine

Having any of the above conditions increases the risk of acquiring meningitis.

TABLE

Causes of Bacterial Meningitis by Route

Route	Bacteria
Infection in or around the head (eg, sinusitis, otitis, mastoiditis), sometimes with a leak of cerebrospinal fluid	<i>Streptococcus pneumoniae</i> <i>Haemophilus influenzae</i> Anaerobic and microaerophilic streptococci <i>Bacteroides</i> species <i>Staphylococcus aureus</i>
Penetrating head wound	<i>S. aureus</i>
Damaged skin (eg, skin infections, abscesses, pressure ulcers, large burns)	<i>S. aureus</i>
An infected shunt	<i>S. epidermidis</i>
A neurosurgical procedure	Gram-negative bacteria (eg, <i>Klebsiella pneumoniae</i> , <i>Acinetobacter calcoaceticus</i> , <i>Escherichia coli</i>)

Immune status

Overall, the most common causes of bacterial meningitis in immunocompromised patients are

- *S. pneumoniae*
- *L. monocytogenes*
- *Pseudomonas aeruginosa*
- *Mycobacterium tuberculosis*
- *N. meningitidis*
- Gram-negative bacteria

But the most likely bacteria depend on the type of immune deficiency:

- **Defects in cell-mediated immunity** (eg, in severe HIV infection, Hodgkin lymphoma, or drug-induced immunosuppression): *L. monocytogenes* or mycobacteria
- **Defects in humoral immunity** or splenectomy: *S. pneumoniae* or, less frequently, *N. meningitidis* (both can cause fulminant meningitis)
- **Neutropenia**: *P. aeruginosa* or gram-negative enteric bacteria

In very young infants (particularly preterm infants) and older adults, T-cell immunity may be weak; thus, these age groups are at risk of meningitis due to *L. monocytogenes*.

Symptoms and Signs of Acute Bacterial Meningitis

In most cases, bacterial meningitis begins with 3 to 5 days of insidiously progressive nonspecific symptoms including malaise, fever, irritability, and vomiting. However, meningitis may be more rapid in onset and can be fulminant, making bacterial meningitis one of the few disorders in which a previously healthy young person may go to sleep with mild symptoms and never awaken.

Typical symptoms and signs of meningitis include

- Fever
- Tachycardia
- Headache
- Photophobia
- Changes in mental status (eg, lethargy, obtundation)
- Nuchal rigidity (although not all patients report it)
- Back pain (less intense than and overshadowed by headache)

However, fever, headache, and nuchal rigidity may be absent in neonates and infants (see [Neonatal Bacterial Meningitis](#)). So-called paradoxical irritability, in which cuddling and consoling by a parent irritates rather than comforts the neonate, suggests bacterial meningitis. If meningitis becomes severe in neonates and infants, skull fontanelles may bulge because of increased intracranial pressure.

Seizures occur early in up to 40% of children with acute bacterial meningitis and may occur in adults. Up to 14% of adult patients present in coma ([1](#)).

Severe meningitis increases intracranial pressure (ICP) and typically causes [papilledema](#), but papilledema may be absent early or be attenuated because of age-related or other factors.

Accompanying systemic infection by the organism may cause

- Rashes, petechiae, or purpura (which suggest meningococcemia)
- Pulmonary consolidation (often in meningitis due to *S. pneumoniae*)
- Heart murmurs (which suggest endocarditis—eg, often caused by *S. aureus* or *S. pneumoniae*)

Atypical presentations in adults

Fever and nuchal rigidity may be absent or mild in immunocompromised or older patients and in patients with alcohol use disorder. Often, in older patients, the only sign is confusion in those who were previously alert or altered responsiveness in those who have dementia. In such patients, as in neonates, the threshold for doing lumbar puncture should be low. Brain imaging (MRI or, less optimally, CT) should be done if focal neurologic deficits are present or increased ICP is suspected.

If bacterial meningitis develops after a neurosurgical procedure, symptoms often take days to develop.

Symptoms and signs reference

1. [van de Beek D, de Gans J, Spanjaard L, et al.](#) Clinical features and prognostic factors in adults with bacterial meningitis. *N Engl J Med*.351(18):1849-1859, 2004. doi: 10.1056/NEJMoa040845. Erratum in: *N Engl J Med*352(9):950, 2005

Diagnosis of Acute Bacterial Meningitis

- Cerebrospinal fluid (CSF) analysis

Bacterial meningitis is a medical emergency; administration of antibiotics should occur within 30 minutes of presentation.

Blood cultures and [lumbar puncture](#) for CSF analysis (unless contraindicated) may be done after initiating antimicrobial therapy because a lumbar puncture within 4 hours of treatment is still likely to yield positive cultures of the etiologic agent. Ideally, blood should be sampled at the time of lumbar puncture so that blood glucose levels can be compared with CSF glucose levels. Treatment should be started as follows:

- If bacterial meningitis is suspected, antibiotics and corticosteroids are given immediately, even before lumbar puncture.
- If bacterial meningitis is suspected and lumbar puncture will be delayed pending CT or MRI, antibiotics and corticosteroids should be started after blood cultures but before neuroimaging is done; the need for confirmation should not delay treatment.

Clinicians should suspect bacterial meningitis in patients with typical symptoms and signs, usually fever, changes in mental status, and nuchal rigidity. However, clinicians must be aware that symptoms and signs are different in neonates and infants and may be absent or initially mild in older patients, in patients with alcohol use disorder, and in immunocompromised patients. Diagnosis can be challenging in the following patients:

- Those who have had a neurosurgical procedure because such procedures can also cause changes in mental status and neck stiffness
- Older patients and patients with alcohol use disorder because changes in mental status may be due to metabolic encephalopathy (which may have multiple causes) or to falls and subdural hematomas

Focal seizures or focal neurologic deficits may indicate a focal lesion such as a [brain abscess](#).

Because untreated bacterial meningitis is lethal, tests should be done if there is even a small chance of meningitis. Testing is particularly helpful in infants, older patients, patients with alcohol use disorder, immunocompromised patients, and patients who had neurosurgical procedure because symptoms may be atypical.

Pearls & Pitfalls

- Do a lumbar puncture even if clinical findings are not specific for meningitis, particularly in infants, older patients, patients with alcohol use disorder, immunocompromised patients, and patients who have had neurosurgery.

If findings suggest acute bacterial meningitis, routine tests include

- CSF analysis
- Complete blood count and differential
- Metabolic panel
- Blood cultures plus polymerase chain reaction (PCR), if available

Lumbar puncture

Unless contraindicated, [lumbar puncture](#) (see also [How To Do Lumbar Puncture](#)) is done immediately to obtain CSF for analysis, the mainstay of diagnosis.

Contraindications to immediate lumbar puncture are signs suggesting markedly increased intracranial pressure (ICP) or an intracranial mass effect (eg, due to edema, hemorrhage, or tumor). Thus, lumbar puncture should be considered high risk with any of the following:

- Papilledema
- Focal neurologic deficits
- Focal seizures
- A known central nervous system mass lesion
- Large stroke
- Suspected central nervous system focal infection

In such cases, lumbar puncture may cause [brain herniation](#) and thus is deferred until neuroimaging (typically CT or MRI) is done to check for increased ICP or a mass effect. When lumbar puncture is deferred, treatment is best begun immediately (after blood sampling for culture and before neuroimaging). After ICP, if increased, has been lowered or if no mass effect or obstructive hydrocephalus is detected, lumbar puncture can be done.

CSF should be sent for analysis: cell count, protein, glucose, Gram staining, culture, PCR, and other tests as indicated clinically. Because *Treponema pallidum* requires long-term *in vitro* culture methods, a CSF VDRL (venereal disease research laboratory) test should be performed ([1](#)). A multiplex film-array PCR panel can provide rapid screening for multiple bacteria and viruses plus *Cryptococcus neoformans* in a CSF sample. This test, which is not always available, is used to supplement, not replace, culture and traditional tests. Simultaneously, a blood sample should be drawn and sent to have the CSF: blood glucose ratio determined. CSF cell count should be determined as soon as possible because white blood cells (WBCs) may adhere to the walls of the collecting tube, resulting in a falsely low cell count; in extremely purulent CSF, WBCs may lyse.

Typical CSF findings in bacterial meningitis include the following (see table [Cerebrospinal Fluid Findings in Meningitis](#)):

- Increased pressure
- Fluid that is often turbid
- A high WBC count (consisting predominantly of polymorphonuclear neutrophils)
- Elevated protein
- A low CSF:blood glucose ratio

A CSF glucose level of ≤ 18 mg/dL or a CSF:blood glucose ratio of < 0.23 strongly suggests bacterial meningitis. In acute bacterial meningitis, an elevated protein level (usually 100 to 500 mg/dL) indicates blood-brain barrier injury.

CSF cell count and protein and glucose levels in patients with acute bacterial meningitis are not always typical. Atypical CSF findings may include (2):

- Normal findings in early stages except for the presence of bacteria
- Predominance of lymphocytes in about 14% of patients, particularly in neonates with gram-negative meningitis, patients with meningitis due to *L. monocytogenes*, and some patients with partially treated bacterial meningitis
- Normal glucose in about 9% of patients
- Normal WBC counts in severely immunosuppressed patients

When initial CSF findings are equivocal, a repeat lumbar puncture 12 to 24 hours later can sometimes clarify the direction in which CSF changes are heading or whether there was a laboratory error.

TABLE

Cerebrospinal Fluid Findings in Meningitis

Condition	Predominant Cell Type*	Protein*	Glucose*	Specific Tests
Normal CSF	All lymphocytes† (0–5 cells/mcL)	< 40 mg/dL	> 50% of blood glucose	None
Bacterial meningitis	Leukocytes (usually PMNs), often greatly increased	Elevated	< 50% of blood glucose (may be extremely low)	Gram staining (yield is high if 10^5 colony-forming units of bacteria/mL are present) Bacterial culture

Condition	Predominant Cell Type*	Protein*	Glucose*	Specific Tests
				Multiplex PCR panel if available
Viral meningitis	Lymphocytes (may be mixed; PMNs and lymphocytes during the first 24–48 hours)	Elevated	Usually normal	Multiplex PCR panel (if available) and/or conventional PCR (to check for enteroviruses or herpes simplex, herpes zoster, or West Nile virus) IgM (to check for West Nile virus or other arboviruses)
Tuberculous meningitis‡	PMNs and lymphocytes (usually mixed pleocytosis)	Elevated	< 50% of blood glucose (may be extremely low)	Acid-fast staining PCR Mycobacterial culture (ideally using a CSF sample of ≥ 30 mL) Interferon-gamma tests of serum and (if available) CSF Xpert MTB/RIF [§]
Fungal meningitis	Usually lymphocytes	Elevated	< 50% of blood glucose (may be extremely low)	Cryptococcal antigen test

Condition	Predominant Cell Type*	Protein*	Glucose*	Specific Tests
				Multiplex PCR panel if available (an adjunctive test, not to replace other tests) Serologic tests for <i>Coccidioides immitis</i> or <i>Histoplasma</i> species antigen especially if patients have recently spent time in an endemic area Fungal culture (ideally using a CSF sample of ≥ 30 mL) India ink (for <i>Cryptococcus</i> sp)

* Changes in cell count, glucose, and protein may be minimal in severely immunocompromised patients. Blood glucose should be measured when lumbar puncture is done so that the CSF:serum glucose ratio can be determined.

† A small number of cells may be present normally in neonates or after a seizure.

‡ In tuberculous meningitis, CSF acid-fast staining can be insensitive, sensitivity of PCR is only about 50%, and culture requires up to 8 weeks. Positive CSF interferon-gamma tests indicate tuberculous meningitis, but serum interferon-gamma tests may only indicate prior infection. Thus, confirming a diagnosis of tuberculous meningitis is difficult, and if it is strongly suspected, even if not confirmed, it is treated presumptively.

§ Xpert MTB/RIF (an automated rapid nucleic acid amplification test) may be used to detect *M. tuberculosis* DNA in CSF.

CSF = cerebrospinal fluid; PCR = polymerase chain reaction; PMNs = polymorphonuclear neutrophils.

Identification of the causative bacteria in CSF involves Gram staining, culture, and, when available, PCR. Gram staining provides information rapidly, but the information is limited. For bacteria to be reliably detected with Gram stain, about 10^5 bacteria/mL must be present. Results may be falsely negative if any of the following occur:

- CSF is handled carelessly.
- Bacteria are not adequately resuspended after CSF has been allowed to settle.
- Errors in decolorization or reading of the slide occur.

If clinicians suspect an anaerobic infection or other unusual bacteria, they should tell the laboratory before samples are plated for cultures. Prior antibiotic therapy can reduce the yield from Gram staining and culture. PCR, if available, and latex agglutination tests to detect bacterial antigens may be useful adjunctive tests, especially in patients who have already received antibiotics.

Determination of antibiotic sensitivity requires bacterial culture.

Until the cause of meningitis is confirmed, other tests using samples of CSF or blood may be done to check for other causes of meningitis, such as viruses (particularly herpes simplex), fungi, and cancer cells.

Other tests

Samples from other sites suspected of being infected (eg, urinary or respiratory tract) should also be cultured.

Diagnosis references

1. [Edmondson DG, Hu B, Norris SJ](#). Long-Term *In Vitro* Culture of the Syphilis Spirochete *Treponema pallidum* subsp. *pallidum*. *mBio*. 2018;9(3):e01153-18. Published 2018 Jun 26. doi:10.1128/mBio.01153-18
2. [de Almeida SM, Furlan SMP, Cretella AMM, et al](#). Comparison of Cerebrospinal Fluid Biomarkers for Differential Diagnosis of Acute Bacterial and Viral Meningitis with Atypical Cerebrospinal Fluid Characteristics. *Med Princ Pract*. 2020;29(3):244-254. doi:10.1159/000501925

Treatment of Acute Bacterial Meningitis

- Antibiotics
- Corticosteroids to decrease cerebral inflammation and edema

Antibiotics are the mainstay of therapy for acute bacterial meningitis. In addition to antibiotics, treatment includes measures to decrease brain and cranial nerve inflammation and increased intracranial pressure (ICP).

Most patients are admitted to an intensive care unit (ICU).

Antibiotics

Antibiotics must be bactericidal for the causative bacteria and must be able to penetrate the blood-brain barrier.

If patients appear ill and findings suggest meningitis, antibiotics (see table [Initial Antibiotics for Acute Bacterial Meningitis](#)) and corticosteroids are started as soon as blood cultures are drawn and even before lumbar puncture. Also, if lumbar puncture is delayed pending neuroimaging results, antibiotic and corticosteroid treatment begins before neuroimaging.

Pearls & Pitfalls

- If patients appear ill and acute meningitis is suspected, treat them with antibiotics and corticosteroids as soon as blood for cultures is drawn.

Appropriate empiric antibiotics depend on the patient's age and immune status and route of infection (see table [Initial Antibiotics for Acute Bacterial Meningitis](#)). In general, clinicians should use antibiotics that are effective against *S. pneumoniae*, *N. meningitidis*, and *S. aureus*. In pregnant women, neonates, older patients, and immunocompromised patients, *Listeria* meningitis is possible; it requires specific antibiotic treatment, usually [ampicillin](#). Herpes simplex encephalitis can clinically mimic early bacterial meningitis; thus, [acyclovir](#) is added. Antibiotic therapy may need to be modified based on results of culture and sensitivity testing.

Commonly used antibiotics include

- Third-generation cephalosporins for *S. pneumoniae* and *N. meningitidis*
- [Ampicillin](#) for *L. monocytogenes*
- [Vancomycin](#) for penicillin-resistant strains of *S. pneumoniae* and for *S. aureus*

For patients who develop meningitis or another infection after a neurosurgical procedure (including shunt implantation), the recommended empiric therapy is [vancomycin](#) plus an antipseudomonal beta-lactam (such as [cefepime](#), [ceftazidime](#), or [meropenem](#)); the choice of empiric beta-lactam agent should be based on local in vitro susceptibility patterns (1).

TABLE

Initial Antibiotics for Acute Bacterial Meningitis

Patient Group	Suspected Bacteria	Provisional Antibiotics
Age		
< 3 months	<i>Streptococcus agalactiae</i>	
	<i>Escherichia coli</i> or other gram-negative bacteria	Ampicillin
	<i>Listeria monocytogenes</i>	plus
	<i>Staphylococcus aureus</i> *	Ceftriaxone or cefotaxime
3 months–18 years	<i>Neisseria meningitidis</i>	
	<i>S. pneumoniae</i>	Cefotaxime or ceftriaxone
	<i>S. aureus</i> *	plus
	<i>Haemophilus influenzae</i> †	Vancomycin
18–50 years	<i>S. pneumoniae</i>	Ceftriaxone or cefotaxime
	<i>N. meningitidis</i>	plus
	<i>S. aureus</i> *	Vancomycin
> 50 years	<i>S. pneumoniae</i>	Ceftriaxone or cefotaxime
	<i>L. monocytogenes</i>	plus
	<i>S. aureus</i>	Ampicillin
	Gram-negative bacteria	plus
	<i>N. meningitidis</i> (unusual in this age group)	Vancomycin
Route		
Sinusitis, otitis, CSF leaks	<i>S. pneumoniae</i> ‡	Vancomycin

<https://www.merckmanuals.com/professional/neurologic-disorders/meningitis/acute-bacterial-meningitis>

* *S. aureus* is an uncommon cause of meningitis except when the route is a penetrating head wound or a neurosurgical procedure. However, it can cause meningitis in all patient groups. Thus, [vancomycin](#) or other antistaphylococcal antibiotics should be given if clinicians think that these bacteria are a possible, even if unlikely, cause.

† *H. influenzae* should be considered in children < 5 years with no record of *H. influenzae* type b conjugate vaccination.

‡ *S. pneumoniae* is the most common causative bacteria in patients with a CSF leak or acute otitis. Such patients may be treated with [vancomycin](#) and [ceftriaxone](#) or [cefotaxime](#). However, when meningitis is accompanied by subdural empyema or develops after a neurosurgical procedure, other bacteria, such as *P. aeruginosa* or *Bacteroides*, may also be present; in such cases, initial treatment should include [vancomycin](#) plus [ceftazidime](#) plus [metronidazole](#). Subdural empyemas should be promptly drained.

CSF = cerebrospinal fluid.

TABLE

Specific Antibiotics for Acute Bacterial Meningitis

Bacteria	Age Group	Antibiotics*	Comments
Gram-positive bacteria (unidentified)	Children and adults	Vancomycin <i>plus</i> Ceftriaxone (cefotaxime) and ampicillin †	—
Gram-negative bacilli (unidentified)	Children and adults	Cefotaxime (or ceftriaxone , meropenem , or ceftazidime) <i>plus</i> Gentamicin , tobramycin , or amikacin ‡ if systemic infection is suspected	—

Bacteria	Age Group	Antibiotics*	Comments
<i>Haemophilus influenzae</i> type b	Children and adults	Ceftriaxone (cefotaxime)	—
<i>Neisseria meningitidis</i>	Children and adults	Ceftriaxone (cefotaxime)	Penicillin G is used for susceptible strains after sensitivities are known.
<i>Streptococcus pneumoniae</i>	Children and adults	Vancomycin plus ceftriaxone or cefotaxime	Penicillin G may be used for susceptible strains after sensitivities are known. Vancomycin covers strains that are highly resistant to penicillin and that ceftriaxone and cefotaxime may not cover. These strains can account for up to 50% of community-acquired infections.
<i>Staphylococcus aureus</i> and <i>S. epidermidis</i>	Children and adults	Vancomycin with or without rifampin	Vancomycin is used for methicillin-resistant strains, or nafcillin or oxacillin may be used after sensitivities are known. Rifampin is added if no improvement occurs with vancomycin or nafcillin .
<i>Listeria</i> species	Children and adults	Ampicillin (penicillin G) or	Penicillin G is used for susceptible strains after

Bacteria	Age Group	Antibiotics*	Comments
		Trimethoprim /sulfa methoxazole	sensitivities are known. Trimethoprim /sulfa methoxazole is used in patients who are allergic to penicillin.
Enteric gram-negative bacteria (eg, <i>Escherichia coli</i> , <i>Klebsiella</i> species, <i>Proteus</i> species)	Children and adults	Ceftriaxone (cefotaxime) <i>plus</i> Gentamicin , tobramycin , or amikacin † if systemic infection is suspected	—
<i>Pseudomonas</i> species	Children and adults	Meropenem (ceftazidime or cefepime), usually alone but sometimes with an aminoglycoside <i>or</i> Aztreonam	—
* Alternative antibiotics are in parentheses. † If gram-positive bacteria are pleomorphic, ampicillin is included to cover <i>Listeria</i> species. ‡ Amikacin is used in areas where gentamicin resistance is common. Because aminoglycosides have poor cerebrospinal fluid penetration, they are infrequently used for treatment of meningitis. When required, they may have to be given intrathecally or via an Ommaya reservoir, especially in patients with <i>Pseudomonas</i> meningitis. When aminoglycosides are used, renal function should be monitored.			

TABLE

Common IV Antibiotic Dosages for Acute Bacterial Meningitis*

Antibiotic	Dosage	
	Children > 1 month	Adults
Ceftriaxone	50 mg/kg every 12 hours	2 g every 12 hours
Cefotaxime	50 mg/kg every 6 hours	2 g every 4–6 hours
Ceftazidime	50 mg/kg every 8 hours	2 g every 8 hours
Cefepime	2 g every 12 hours	2 g every 8–12 hours
Ampicillin	75 mg/kg every 6 hours	2–3 g every 4 hours
Penicillin G	4 million units every 4 hours	4 million units every 4 hours
Nafcillin and oxacillin	50 mg/kg every 6 hours	2 g every 4 hours
Vancomycin †	15 mg/kg every 6 hours	10–15 mg/kg every 8 hours
Meropenem	40 mg/kg every 8 hours	2 g every 8 hours
Gentamicin and tobramycin †	2.5 mg/kg every 8 hours	2 mg/kg every 8 hours
Amikacin †	10 mg/kg every 8 hours	7.5 mg/kg every 12 hours
Rifampin	6.7 mg/kg every 8 hours	600 mg every 24 hours
* For neonatal dosages, see table Recommended Dosages of Selected Parenteral Antibiotics for Neonates.		
† Renal function should be monitored.		

Corticosteroids

[Dexamethasone](#) is used to decrease cerebral and cranial nerve inflammation and edema; it should be given when therapy is started. Adults are given 10 mg IV; children are given 0.15 mg/kg IV.

[Dexamethasone](#) is given immediately before or with the initial dose of antibiotics and every 6 hours for 4 days.

Use of [dexamethasone](#) is best-established for patients with pneumococcal meningitis.

Other measures

The effectiveness of other measures is less well-proved.

Patients presenting with papilledema or signs of impending brain herniation are treated for increased ICP with the following:

- Elevation of the head of the bed to 30°
- Hyperventilation to a PCO₂ of 27 to 30 mm Hg for not more than 24 hours to cause intracranial vasoconstriction
- Osmotic diuresis with IV [mannitol](#)

Hyperventilation is used until other measures become effective and is not used for more than 24 hours. When stopped, the PCO₂ should be gradually increased to normal because a sudden increase may cause a significant increase in ICP.

Usually, adults are given intravenous [mannitol](#) to cause an osmotic diuresis and reduce brain swelling.

Additional measures can include

- IV fluids
- Antiseizure medications
- Treatment of concomitant infections
- Treatment of specific complications (eg, corticosteroids for Waterhouse-Friderichsen syndrome, surgical drainage for subdural empyema)

Treatment reference

1. [Tunkel AR, Hasbun R, Bhimraj A, et al.](#) 2017 Infectious Diseases Society of America's Clinical Practice Guidelines for Healthcare-Associated Ventriculitis and Meningitis. *Clin Infect Dis*. 2017;64(6):e34-e65. doi:10.1093/cid/ciw861

Prognosis for Acute Bacterial Meningitis

With antibiotic treatment, the mortality rate for children < 19 years may be as low as 3% but is often higher; survivors may be deaf and neuropsychologically impaired. The mortality rate, even with antibiotic treatment, is about 21% for adults ([1](#)). Community-acquired meningitis due to *S. aureus* has a mortality rate in the range of 43%.

In general, mortality rate correlates with depth of obtundation or coma. Factors associated with a poor prognosis include

- Age > 60 years
- Coexisting debilitating disorders
- A low Glasgow coma score at admission (see tables [Glasgow Coma Scale](#) and [Modified Glasgow Coma Scale for Infants and Children](#))
- Focal neurologic deficits
- A low CSF cell count

- Increased CSF pressure (particularly)

Seizures and a low CSF:serum glucose ratio may also indicate a poor prognosis.

CLINICAL CALCULATORS

[Glasgow Coma Scale](#)



Prognosis reference

1. [van de Beek D, de Gans J, Spanjaard L, et al](#): Clinical features and prognostic factors in adults with bacterial meningitis [published correction appears in *N Engl J Med*. 2005 Mar 3;352(9):950]. *N Engl J Med*. 2004;351(18):1849-1859. doi:10.1056/NEJMoa040845

Prevention of Acute Bacterial Meningitis

Vaccination

Use of vaccines for *H. influenzae* type B and, to a lesser extent, for *N. meningitidis* and *S. pneumoniae* has reduced the incidence of bacterial meningitis. (See [Haemophilus influenzae Type b \(Hib\) Vaccine](#), [Meningococcal Vaccine](#), and [Pneumococcal Vaccine](#).)

During a meningitis epidemic, the population at risk (eg, college students, a small town) must be identified, and its size must be determined before proceeding to mass vaccination. The effort is expensive and requires public education and support, but it saves lives and reduces morbidity.

Physical measures

Keeping patients in respiratory isolation (using droplet precautions) for the first 24 hours of therapy can help prevent meningitis from spreading. Gloves, masks, and gowns are used.

Chemoprophylaxis

Anyone who has prolonged face-to-face contact with a patient who has meningitis (eg, household or day care contacts, medical personnel and other people who are exposed to the patient's oral secretions) should be given postexposure chemoprophylaxis.

For **meningococcal meningitis**, chemoprophylaxis consists of 1 of the following:

- [Rifampin](#) 600 mg (for children > 1 month, 10 mg/kg; for children < 1 month, 5 mg/kg) orally every 12 hours for 4 doses
- [Ceftriaxone](#) 250 mg (for children < 15 years, 125 mg) IM for 1 dose
- For adults, a fluoroquinolone ([ciprofloxacin](#) or [levofloxacin](#) 500 mg or [ofloxacin](#) 400 mg) orally for 1 dose

For **meningitis due to *H. influenzae* type b**, chemoprophylaxis is [rifampin](#) 20 mg/kg orally once a day (maximum: 600 mg/day) for 4 days. There is no consensus on whether children < 2 years require prophylaxis for exposure at day care.

Chemoprophylaxis is not usually needed for contacts of patients with other types of bacterial meningitis.

Key Points

- Common causes of acute bacterial meningitis include *N. meningitidis* and *S. pneumoniae* in children and adults and *Listeria* species in infants and older adults; *S. aureus* occasionally causes meningitis in people of all ages.
- Typical features may be absent or subtle in infants, patients with alcohol use disorder, older patients, immunocompromised patients, and patients who develop meningitis after a neurosurgical procedure.
- If patients have focal neurologic deficits, obtundation, seizures, or papilledema (suggesting increased ICP or an intracranial mass effect), defer lumbar puncture pending results of neuroimaging.
- Treat acute bacterial meningitis as soon as possible, even before the diagnosis is confirmed.
- Common empirically chosen antibiotic regimens often include third-generation cephalosporins (for *S. pneumoniae* and *N. meningitidis*), [ampicillin](#) (for *L. monocytogenes*), and [vancomycin](#) (for penicillin-resistant strains of *S. pneumoniae* and for *S. aureus*).
- Routine vaccination for *H. influenza*, *S. pneumoniae*, and *N. meningitidis* and chemoprophylaxis against *N. meningitidis* help prevent meningitis.

More Information

The following English-language resources may be useful. Please note that The Manual is not responsible for the content of these resources.

[Practice Guidelines for the Management of Bacterial Meningitis](#): These guidelines provide recommendations for the diagnosis and management of bacterial meningitis, including the initial approach, indications for CT before lumbar puncture, tests to distinguish viral from bacterial meningitis, specific tests to identify the causative bacteria, timing of antimicrobial medications for suspected meningitis, specific medications to be used to treat suspected or confirmed bacterial meningitis, and the role of dexamethasone.

[2017 Infectious Diseases Society of America's Clinical Practice Guidelines for Healthcare-Associated Ventriculitis and Meningitis](#): This guideline reviews the literature, evaluates the evidence, and presents recommendations. It specifically discusses the approach to infections associated with

cerebrospinal fluid shunts, cerebrospinal fluid drains, intrathecal medications (eg, baclofen), deep brain stimulation hardware, neurosurgery, and head trauma.

Drug Information for the Topic



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